

PDF document (pg8, pg19)

Freq band	RX Side				TX Side									
	LR1	LR2	CR1	CR2	Lchoke	L0	C0	CM	LM1	CM1	LM2	CM2	LM3	CM3
390 MHz	75 nH	68 nH	5.6 pF	2.4 pF	270 nH	36 nH	10 pF	18 pF	27 nH	12 pF	22 nH	12 pF	18 nH	12 pF
460 MHz	56 nH	51 nH	3.9 pF	2.2 pF	220 nH	30 nH	8.2 pF	13 pF	27 nH	9.1 pF	22 nH	9.1 pF	15 nH	9.1 pF
490 MHz	51 nH	47 nH	3.9 pF	2.2 pF	220 nH	24 nH	3.9 pF	24 nH	27 nH	5.6 pF	27 nH	9.1 pF	15 nH	9.1 pF
868 MHz	20 nH	24 nH	3.0 pF	1.0 pF	120 nH	15 nH	3.9 pF	5.1 pF	11 nH	2.7 pF	11 nH	4.7 pF	9.1 nH	4.7 pF
915 MHz	18 nH	22 nH	3.0 pF	1.0 pF	100 nH	12 nH	3.9 pF	5.6 pF	10 nH	2.7 pF	10 nH	4.3 pF	8.2 nH	4.3 pF

Freq band	RX Side				TX Side									
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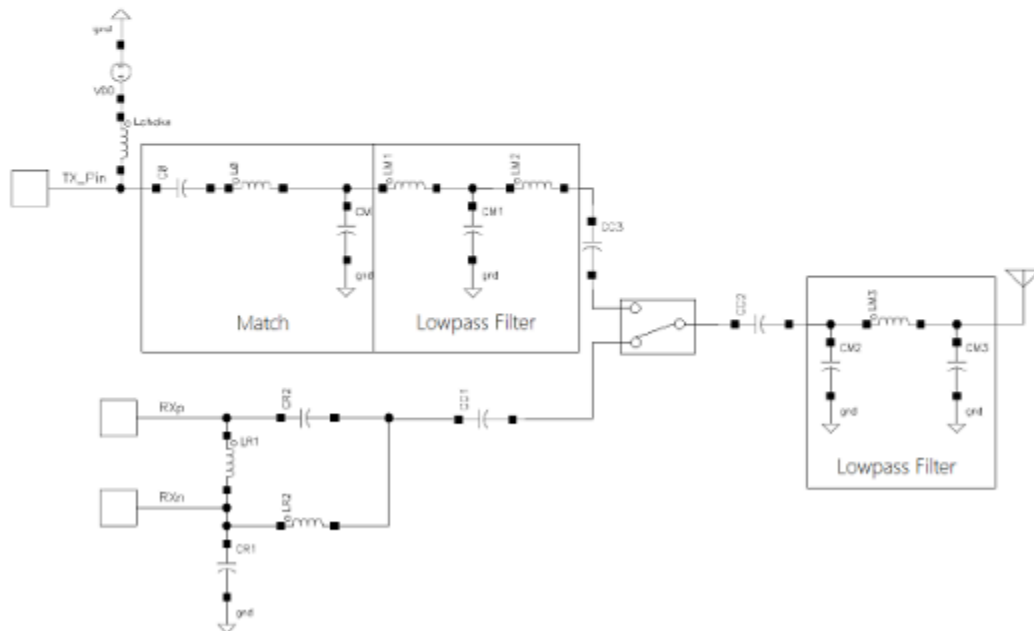
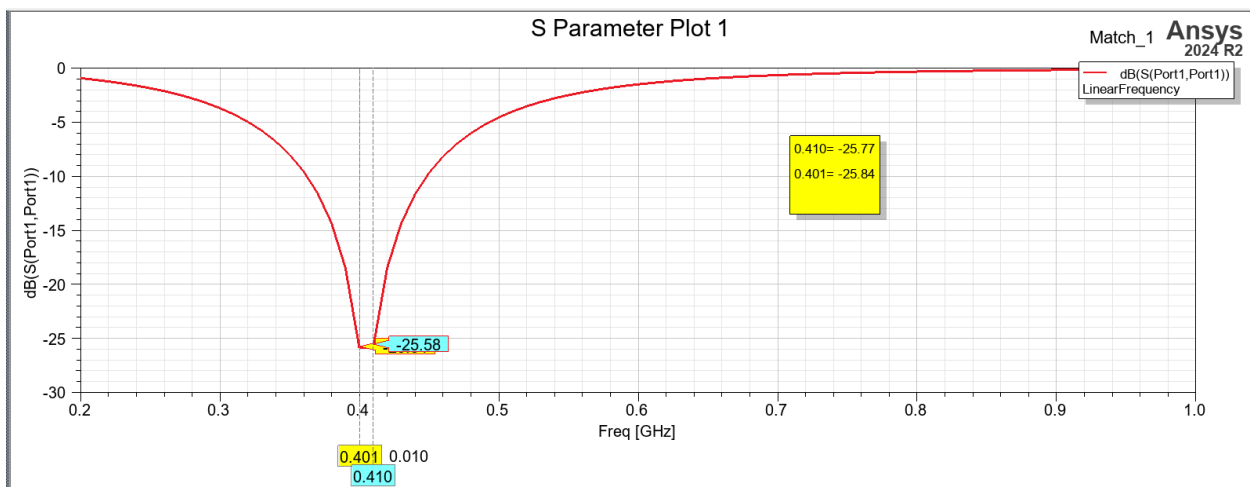
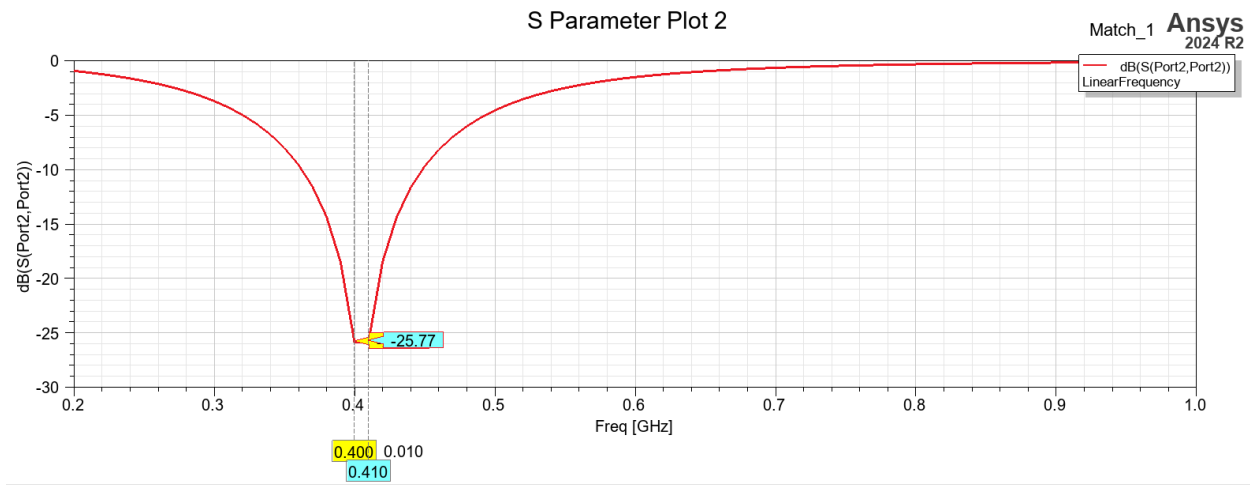
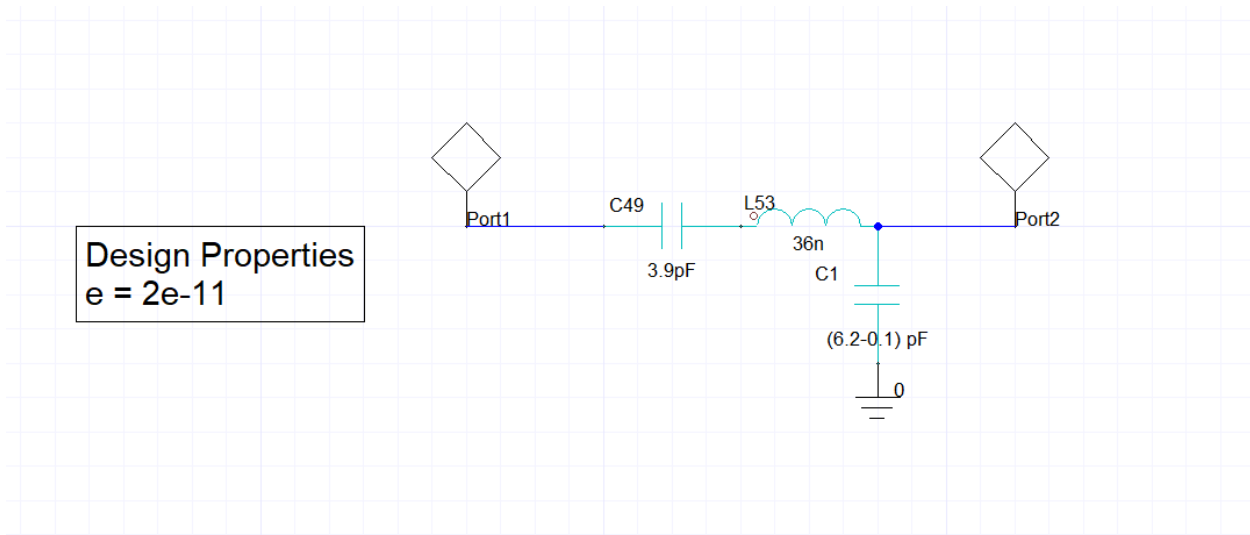
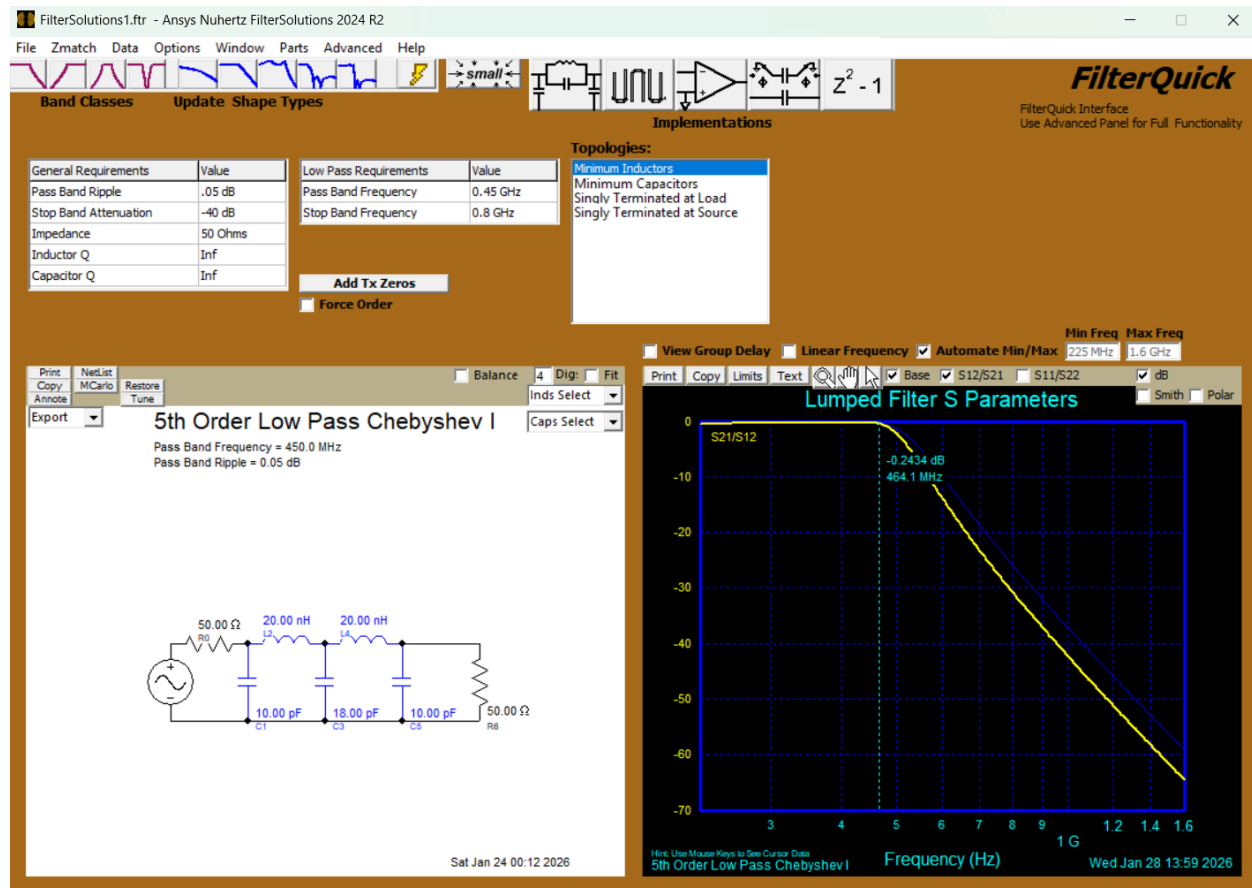


Figure 3. Class-E Matching Topology for Switched TX/RX Board Configuration

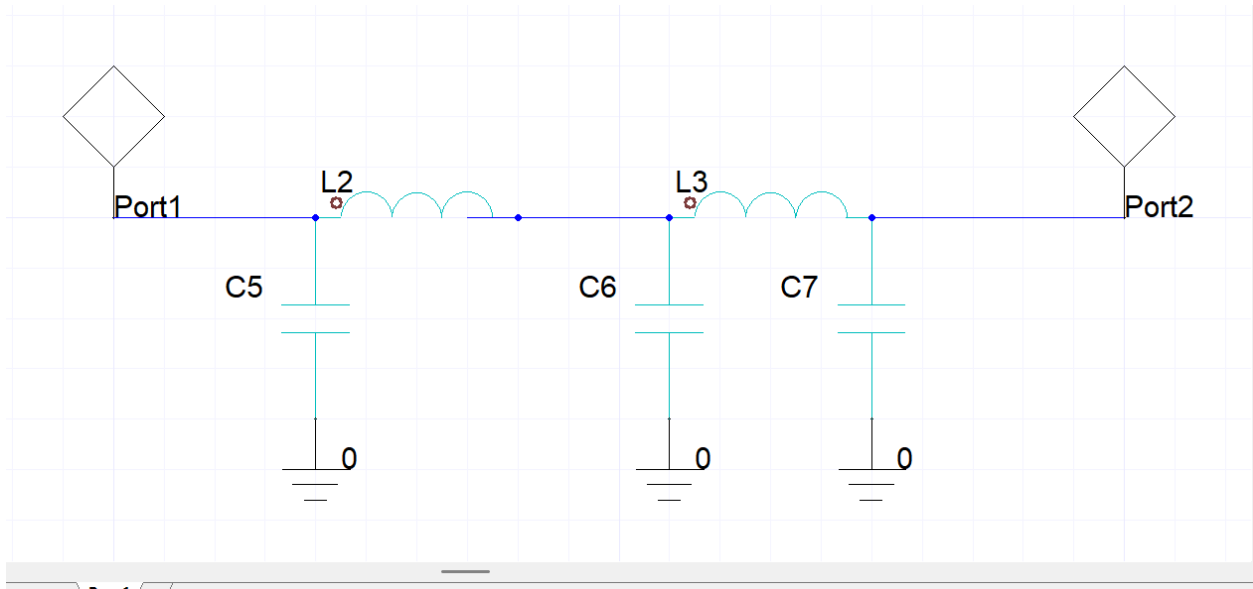
Match Circuit 1-



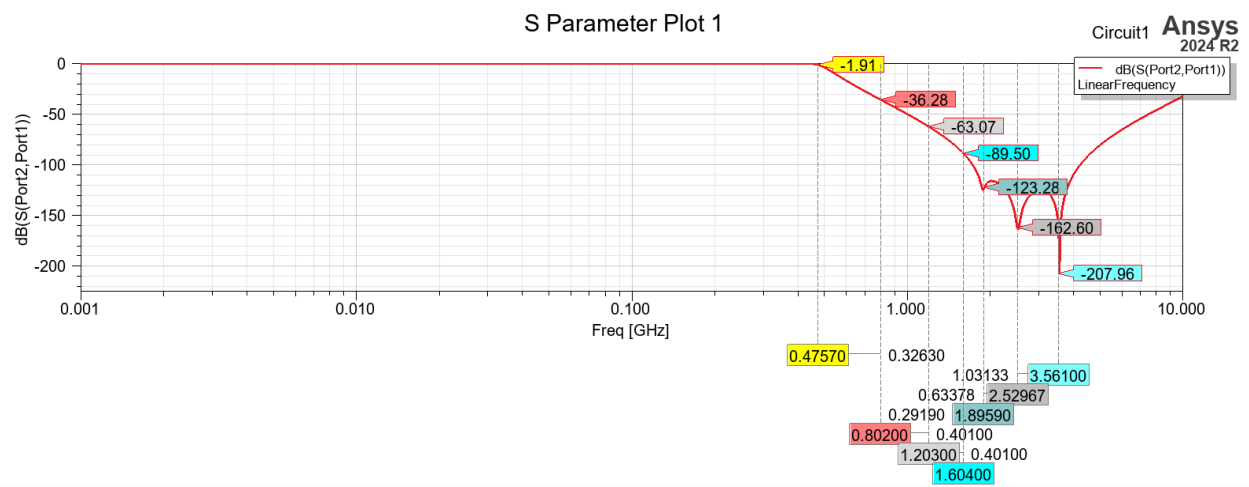
LPF Simulations:



The cutoff is kept much higher than 401 MHz so that the 401 MHz signal stays in the flat part of the filter where it is not weakened. A wide transition band is used on purpose, so even if the filter response shifts slightly because of real-world issues like component tolerance, temperature, or PCB layout, the required signal is still passed cleanly. In simple terms, the cutoff is placed higher to give safety margin, so the useful frequency is never close to the edge of the filter where losses and distortion begin. (also with 5th order chebyshev it introduces 0.05dB ripples and also increases the transition width but the stop band rejection is improved, also checked the filter response in ansys by taking a tolerance of $\pm 5\%$)



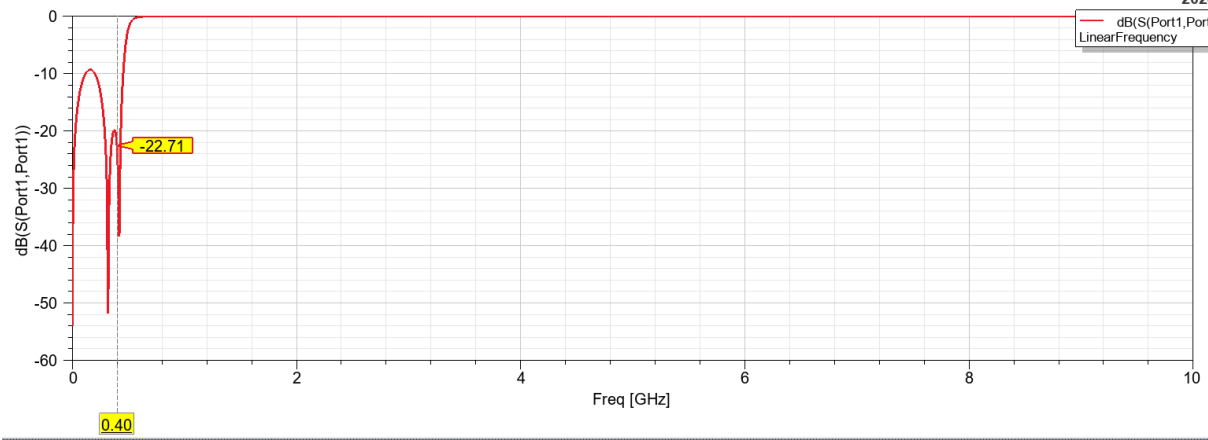
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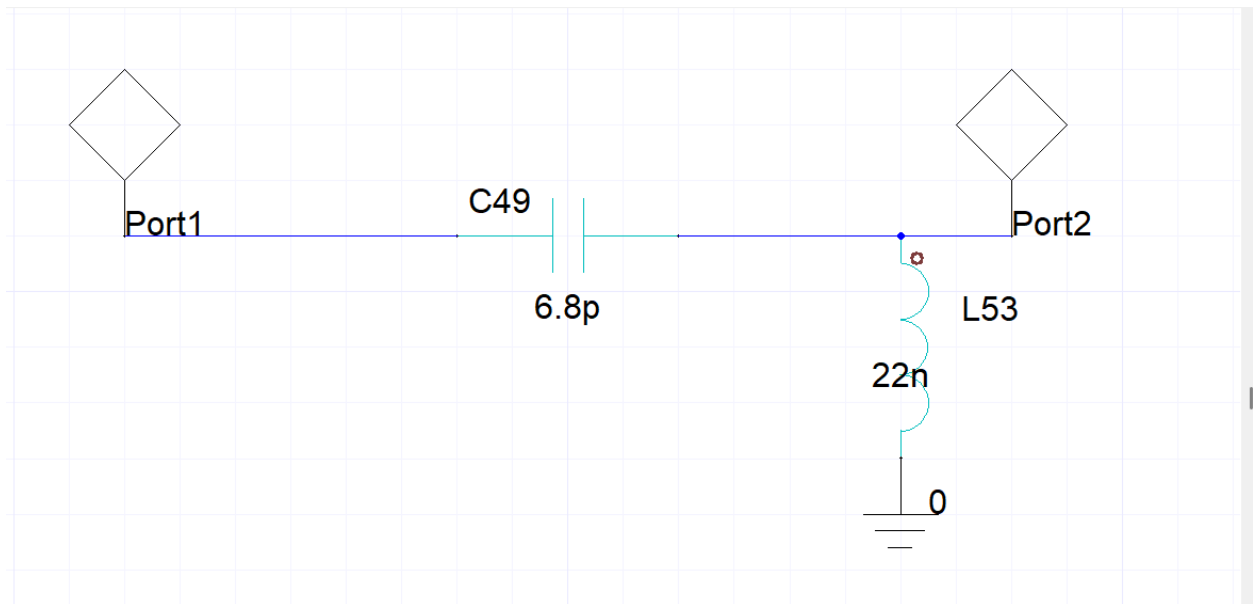
(checked the rejection up to 4th resonant frequency)

S Parameter Plot 2

Circuit1
2024 R2
dB(S(Port1,Port1))
LinearFrequency

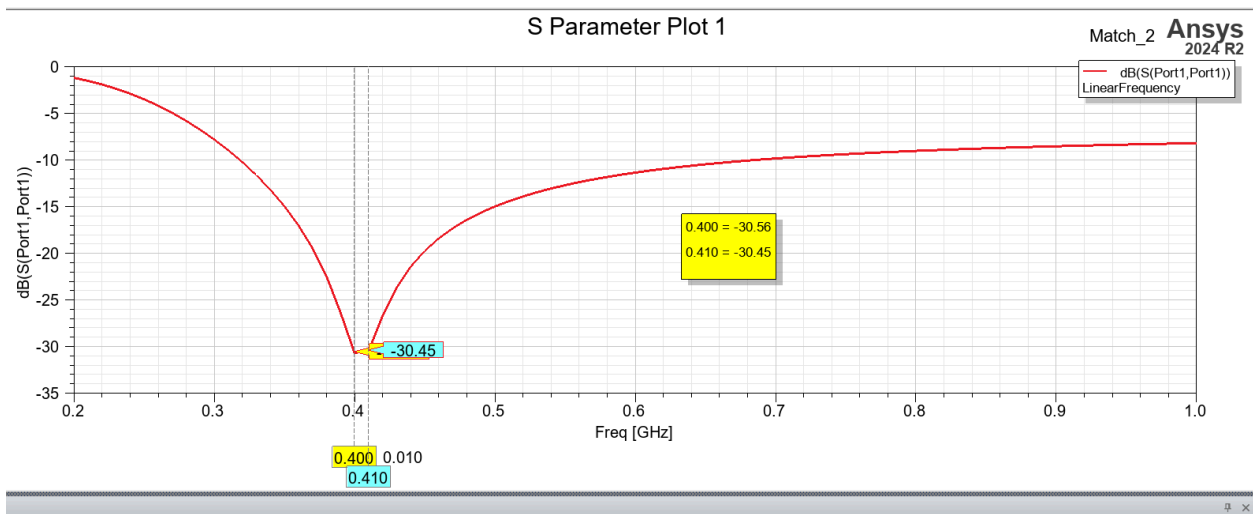


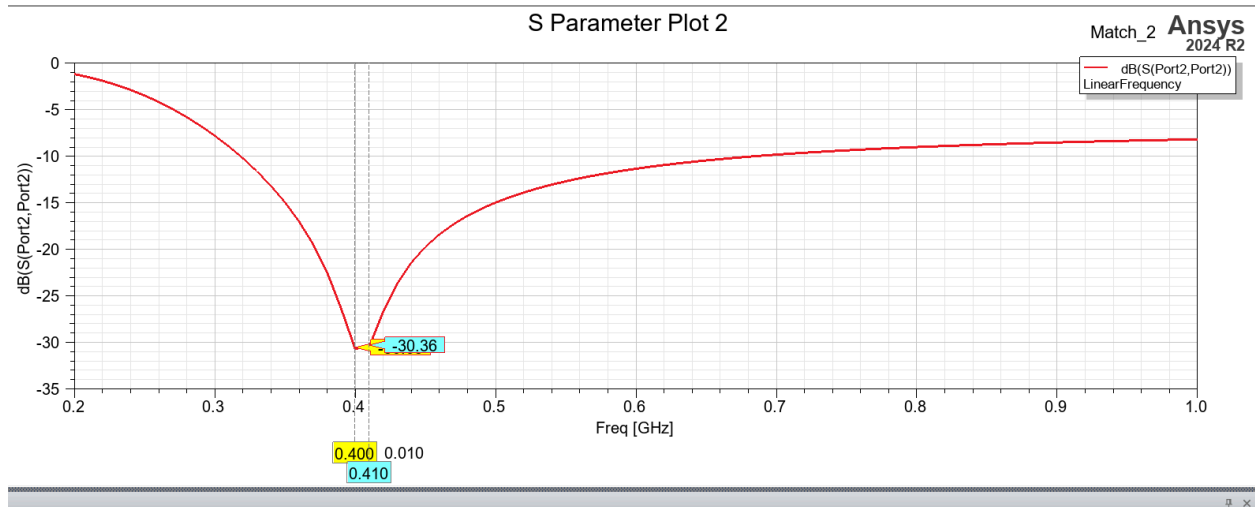
Match 2(Swapped the LC components):



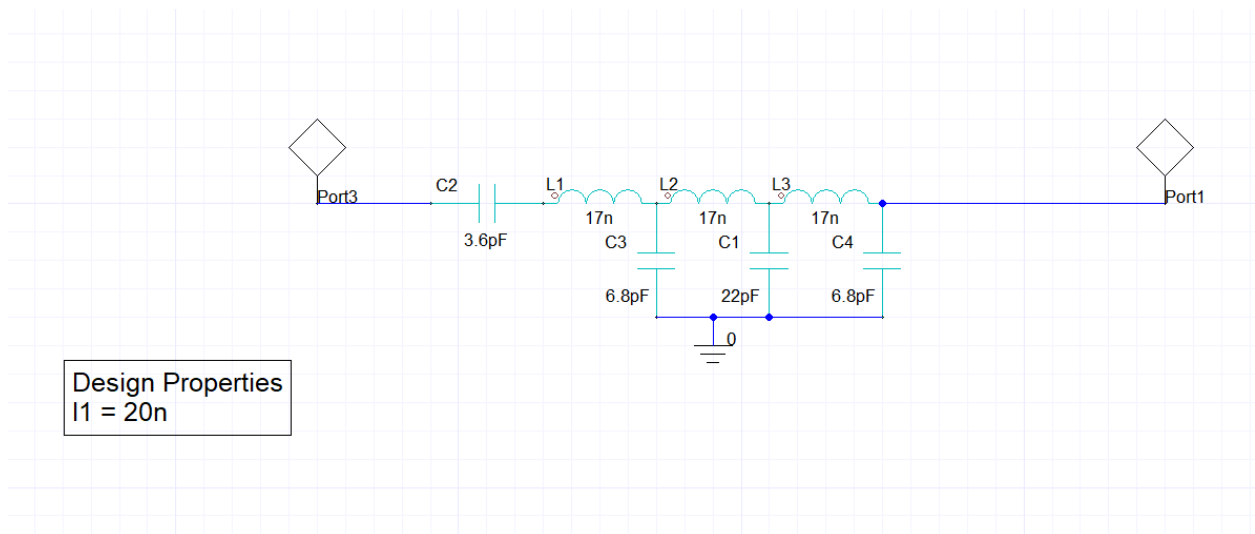
Reference: [Why Your LC Filter Fails \(Even If the Schematic Looks Right\)](#)

Results:

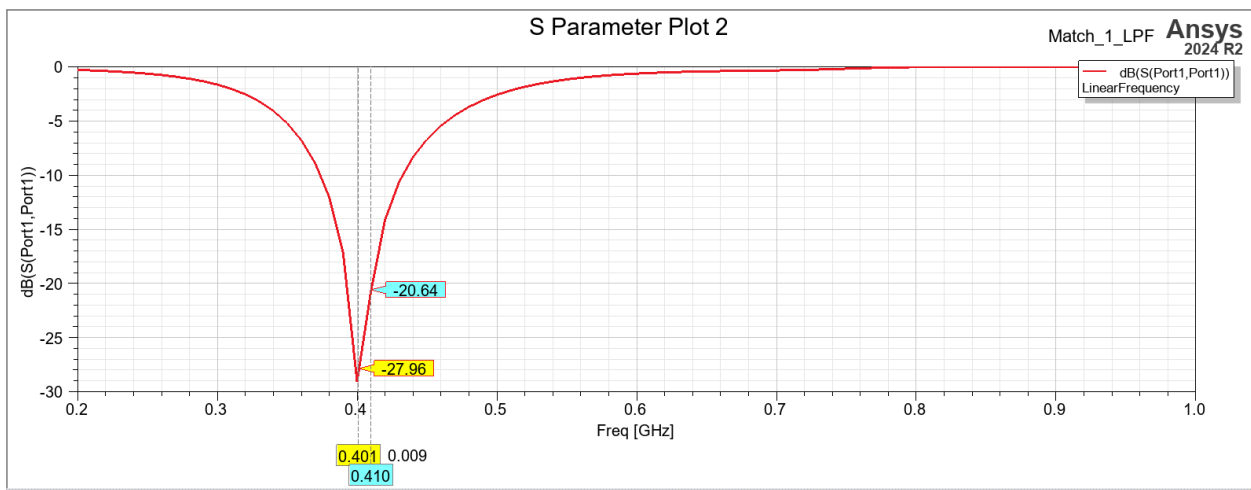
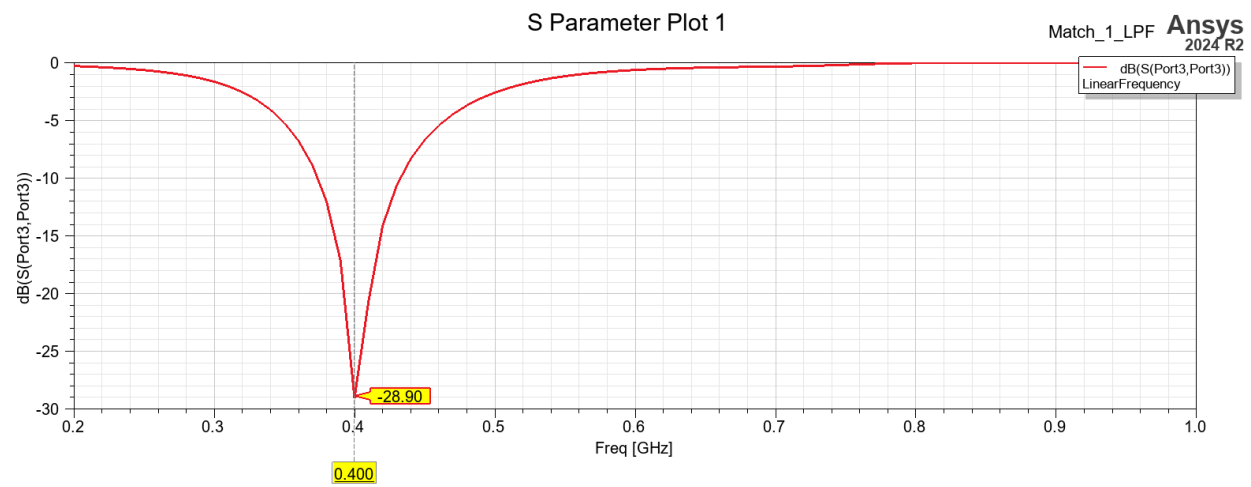




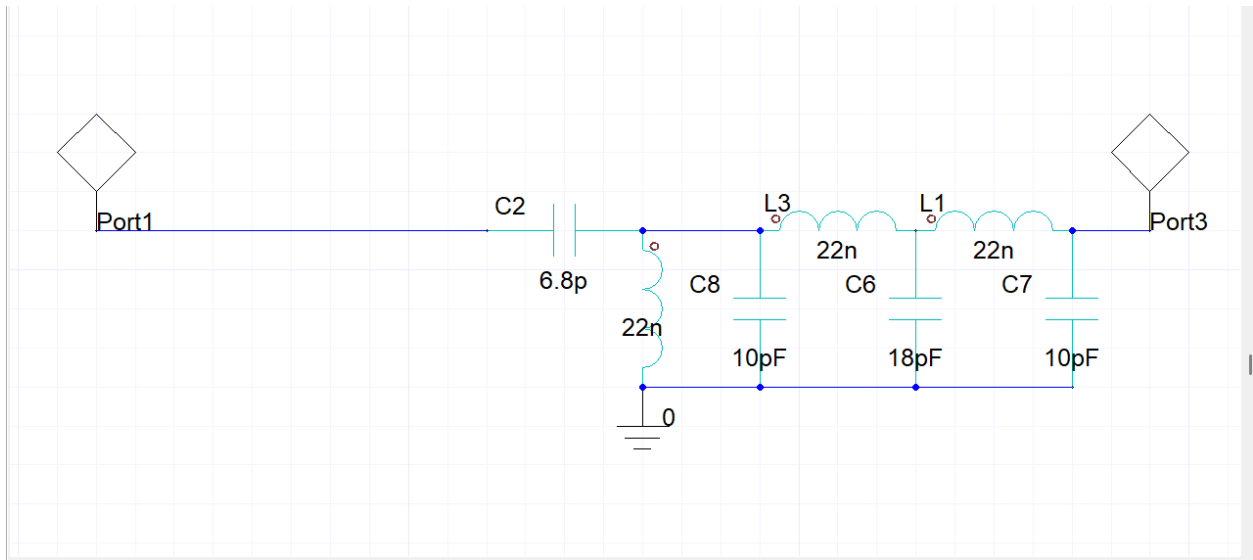
Circuit 1(Match+LPF):



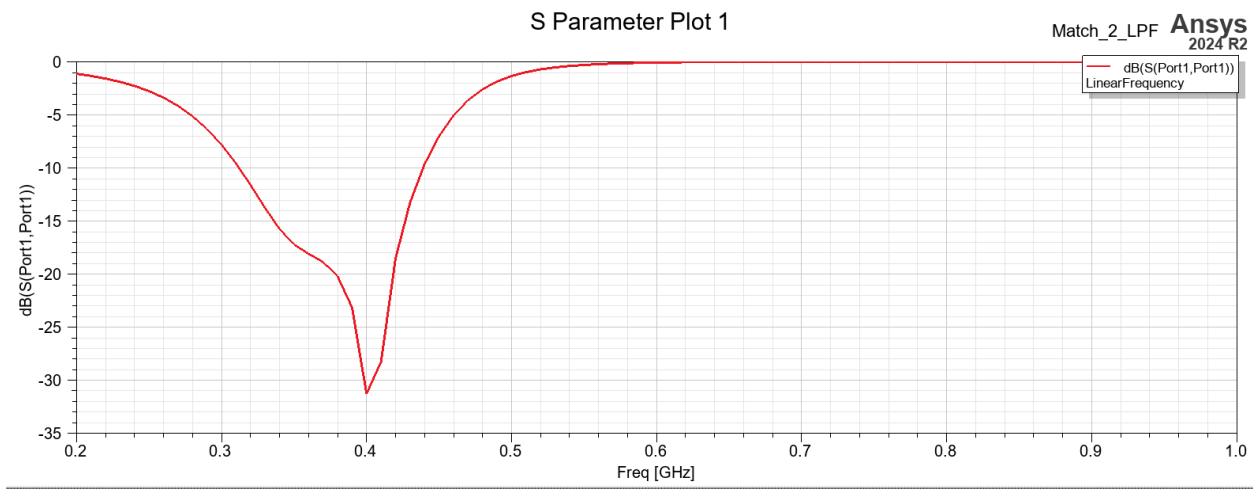
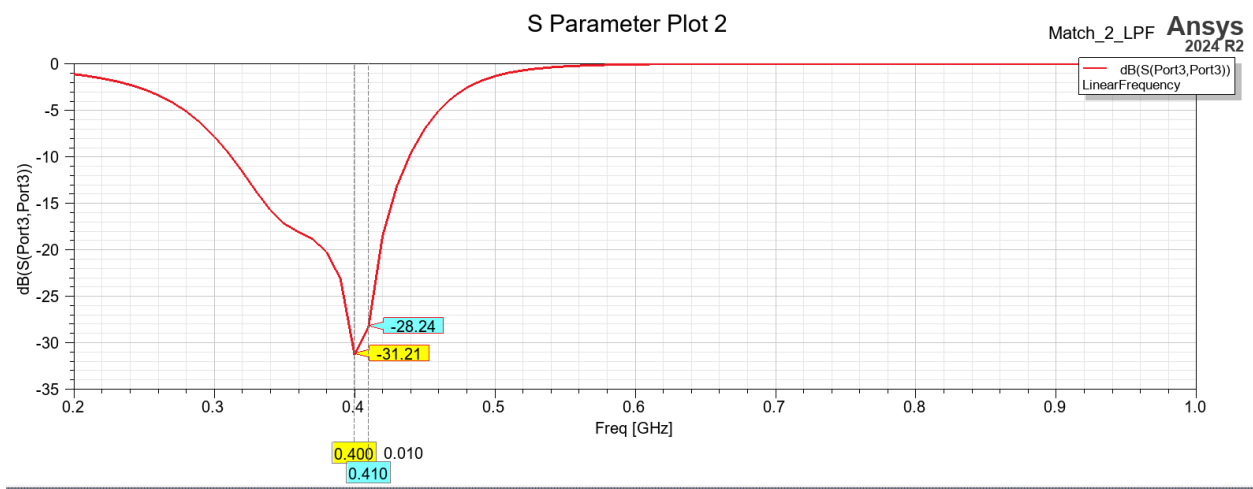
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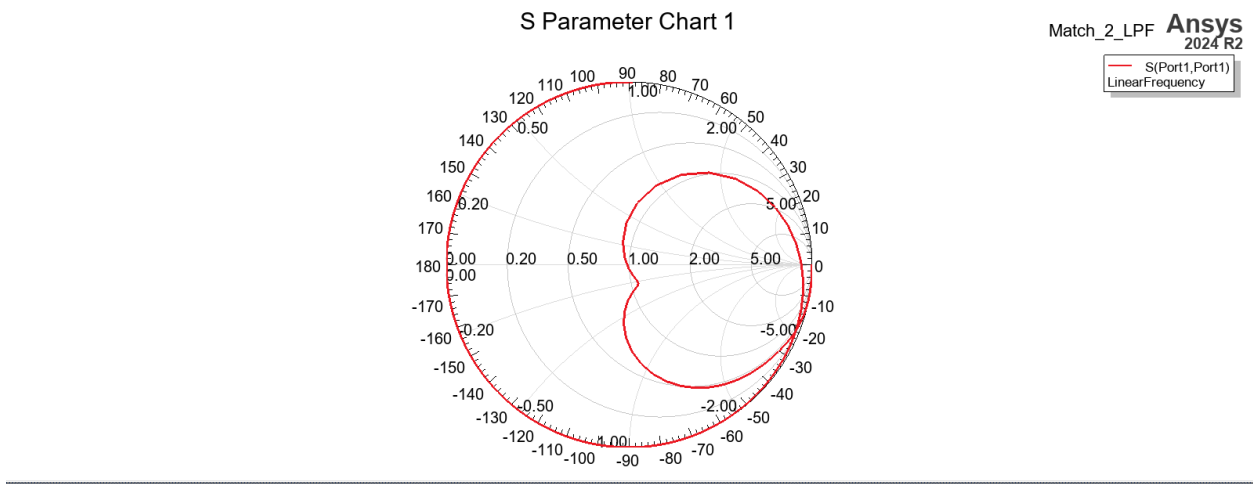
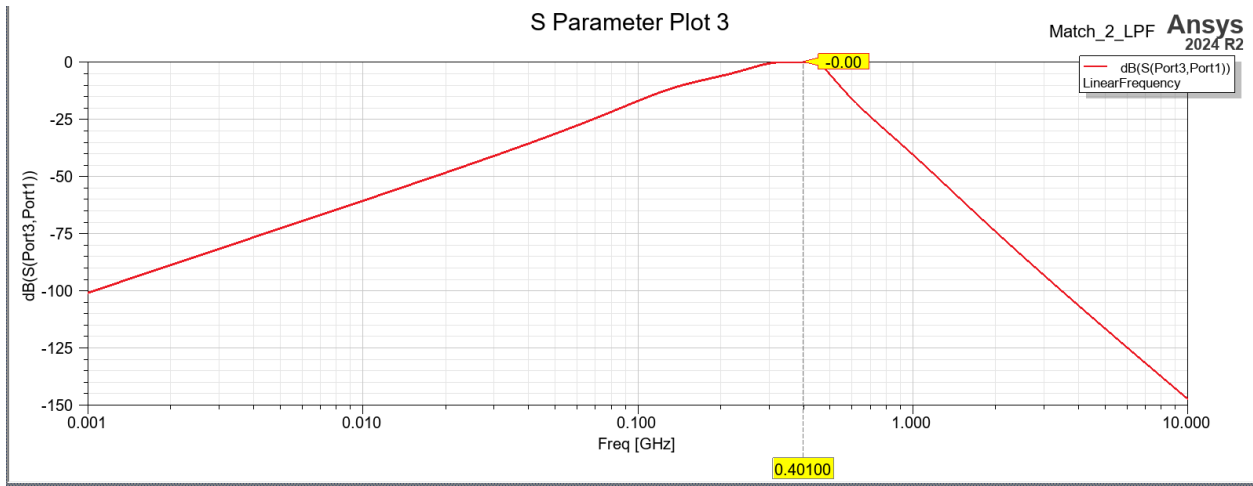


Match_2_Ipf(Match_2+LPF circuit):



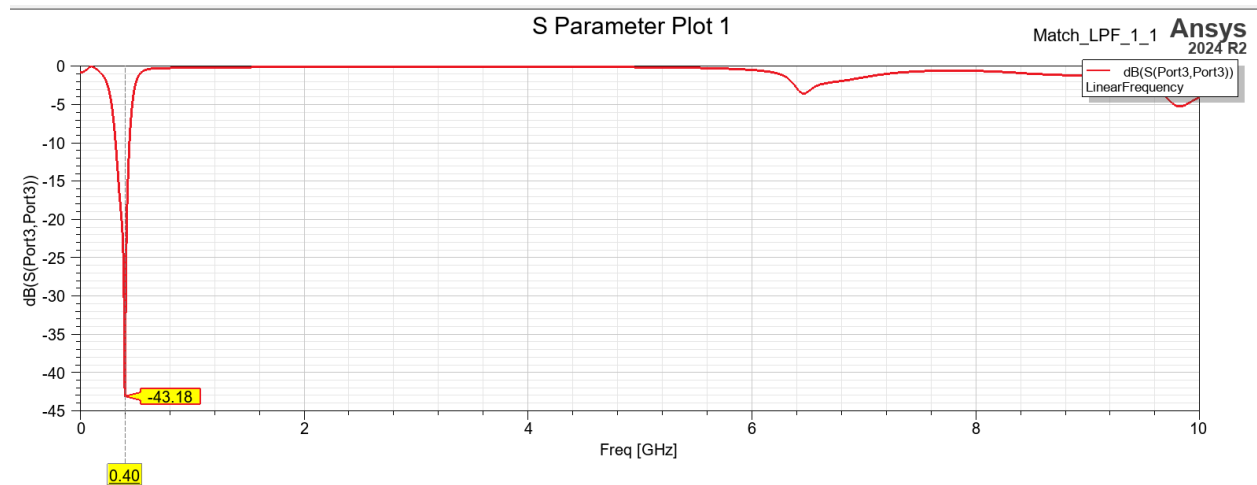
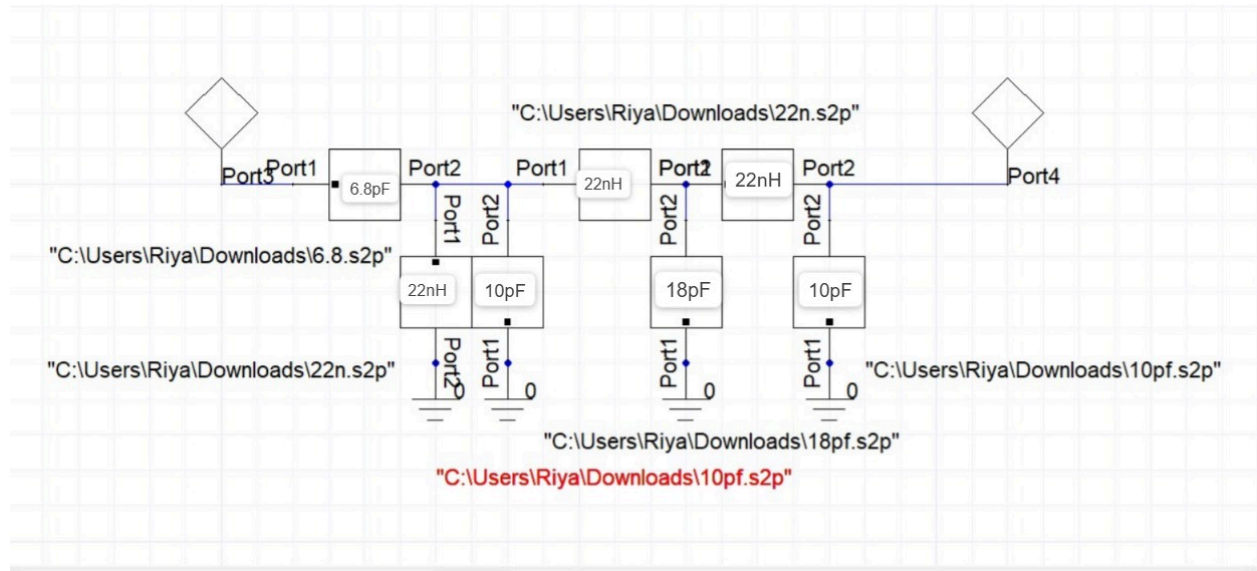
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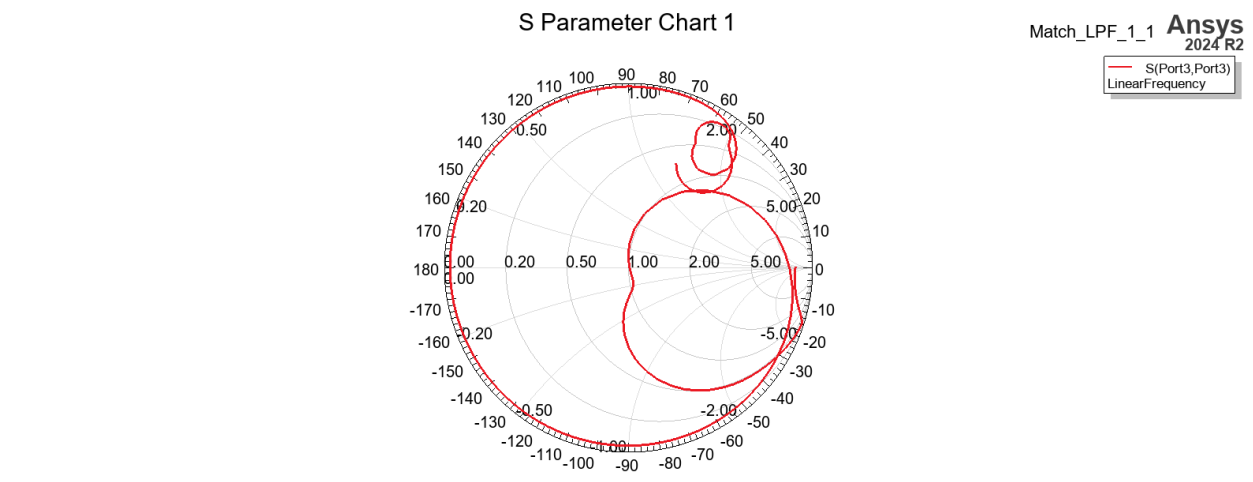
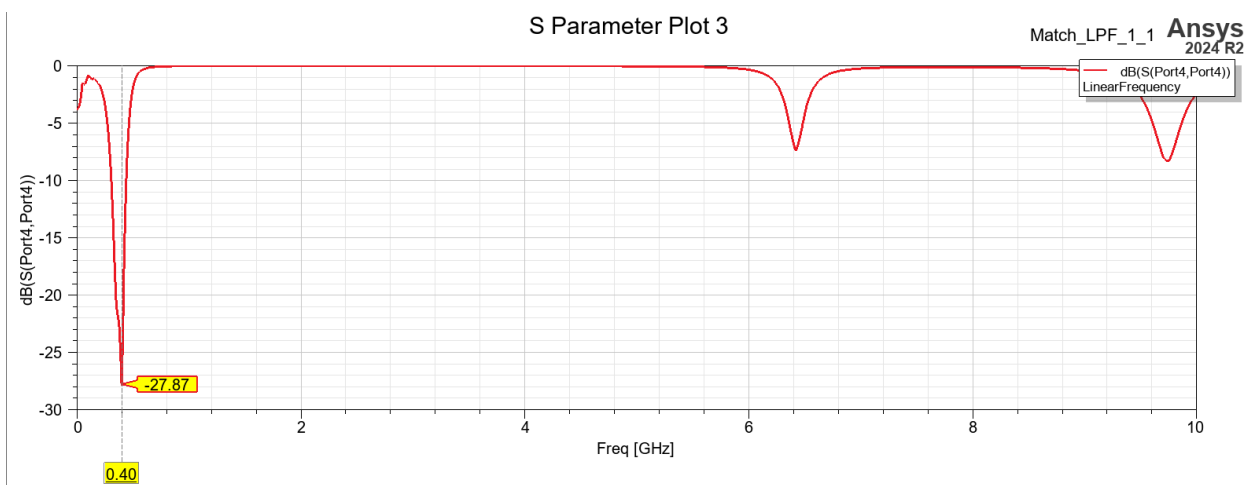
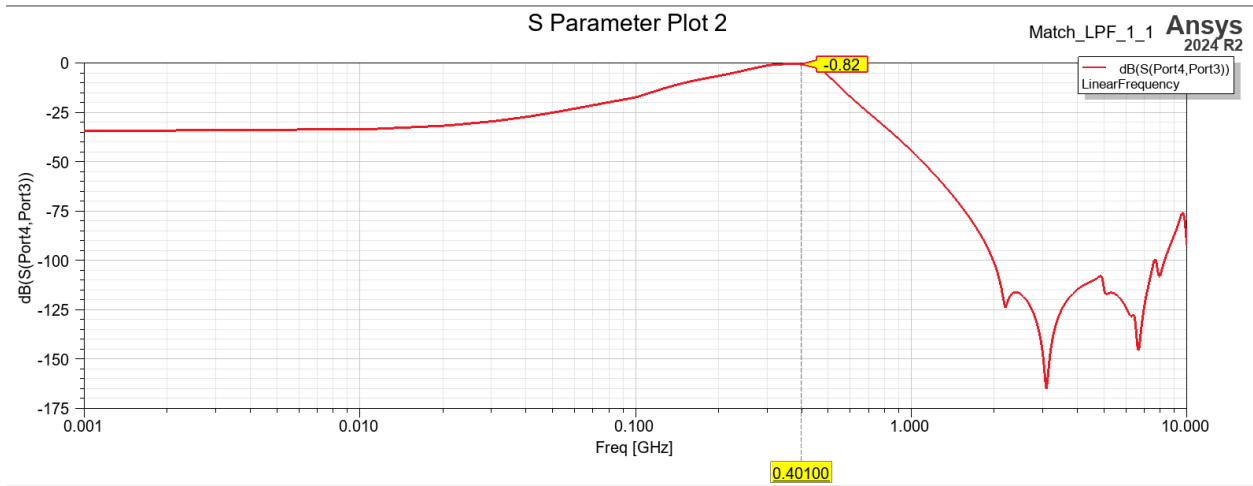




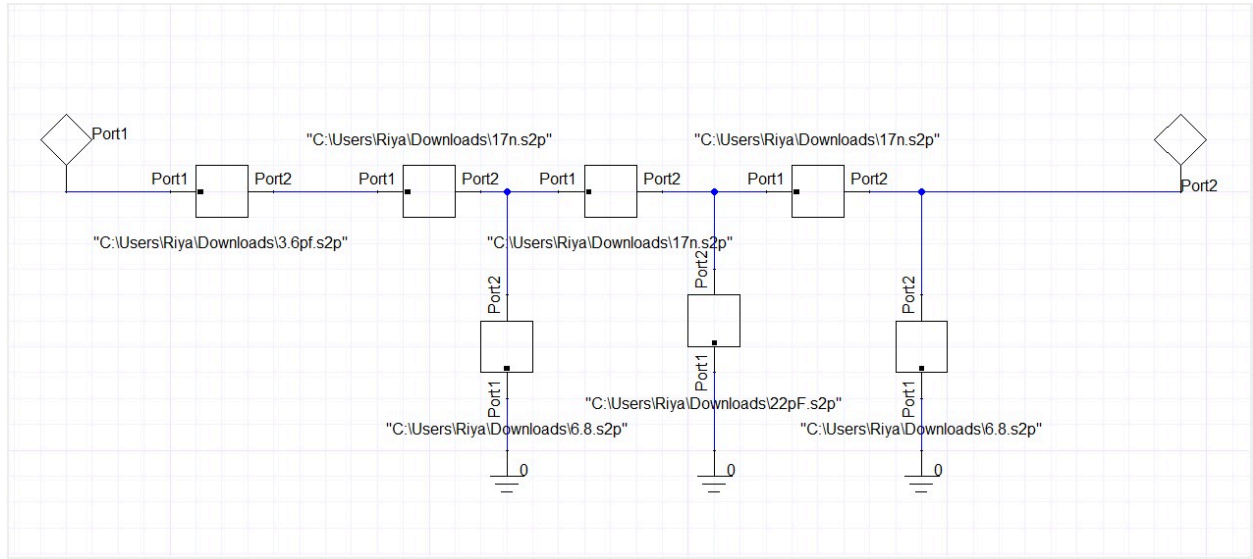
Match 2+LPF(simulated with simsurf components)

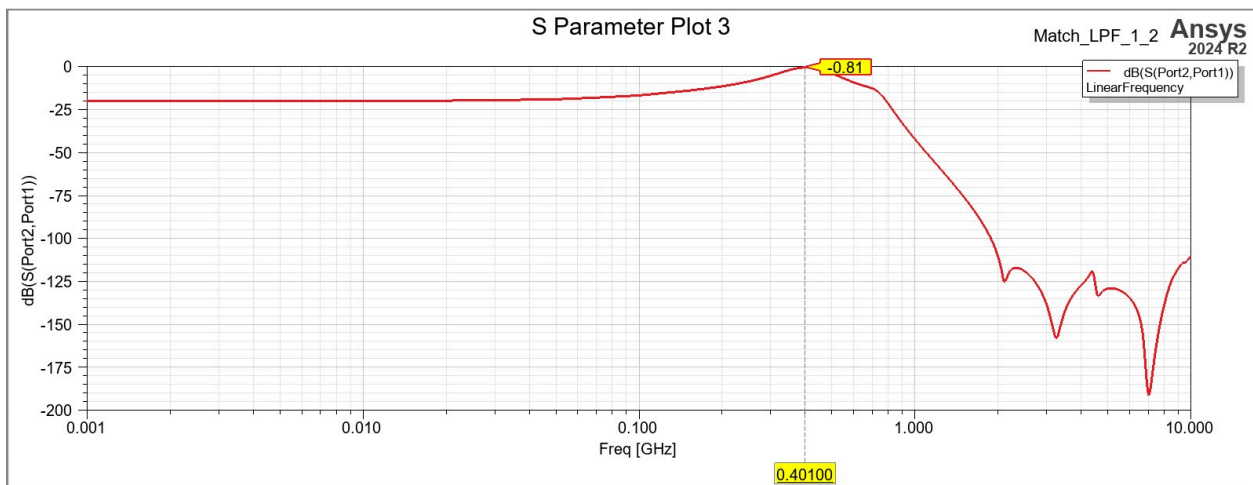
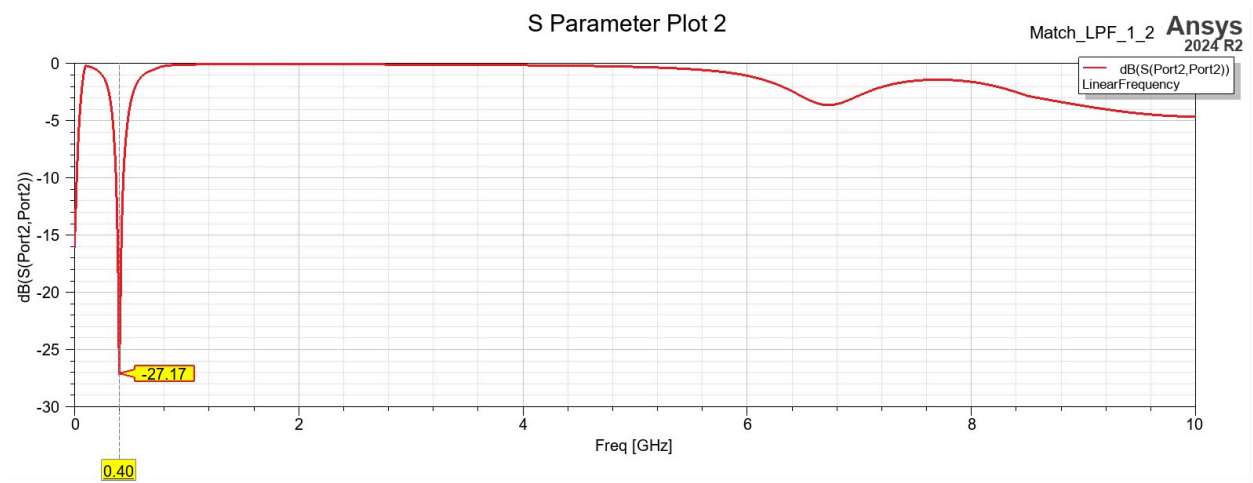
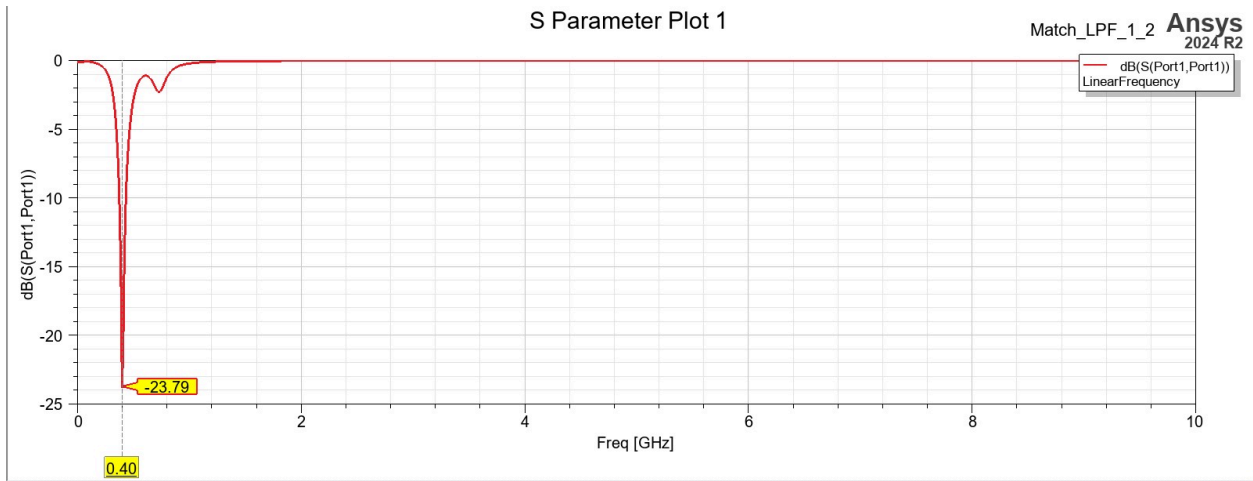
RF Components





Match 1+LPF(simulated with simsurf components)

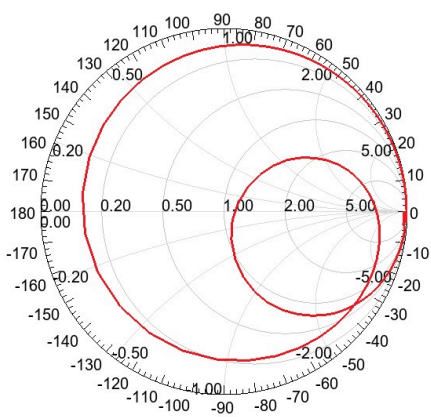




S-Parameters

Match_LPF_1_2 **Ansys**
2024 R2

S(Port1,Port1)
LinearFrequency



RX PATH:

- PDF document
- PDF document

AN648

2.3.3. Component Values for Class-E Switched TX/RX Board Configuration

Table 5 provides the component values required for a Class-E match using the Switched TX/RX board configuration and a supply voltage of VDD = 3.3 V. The 490 MHz matching network includes parallel inductor in the place of CM, instead of a capacitor.

Table 5. Class-E Match Component Values vs. Frequency (Switched TX/RX Board)

Freq band	RX Side				TX Side									
	LR1	LR2	CR1	CR2	Lchoke	L0	C0	CM	LM1	CM1	LM2	CM2	LM3	CM3
390 MHz	75 nH	68 nH	5.6 pF	2.4 pF	270 nH	36 nH	10 pF	18 pF	27 nH	12 pF	22 nH	12 pF	18 nH	12 pF
460 MHz	56 nH	51 nH	3.9 pF	2.2 pF	220 nH	30 nH	8.2 pF	13 pF	27 nH	9.1 pF	22 nH	9.1 pF	15 nH	9.1 pF
490 MHz	51 nH	47 nH	3.9 pF	2.2 pF	220 nH	24 nH	3.9 pF	24 nH	27 nH	5.6 pF	27 nH	9.1 pF	15 nH	9.1 pF
868 MHz	20 nH	24 nH	3.0 pF	1.0 pF	120 nH	15 nH	3.9 pF	5.1 pF	11 nH	2.7 pF	11 nH	4.7 pF	9.1 nH	4.7 pF
915 MHz	18 nH	22 nH	3.0 pF	1.0 pF	100 nH	12 nH	3.9 pF	5.6 pF	10 nH	2.7 pF	10 nH	4.3 pF	8.2 nH	4.3 pF

Table 6. EZR32 R63/68 Class-E Match Component Values vs Frequency (Switched TX/RX Board)

Freq band	RX Side				TX Side									
	LR1	LR2	CR1	CR2	Lchoke	L0	C0	CM	LM1	CM1	LM2	CM2	LM3	CM3
915 MHz	18 nH	22 nH	3.0 pF	1.0 pF	100 nH	11 nH	5.1 pF	4.7 pF	10 nH	2.7 pF	10 nH	4.3 pF	8.2 nH	4.3 pF

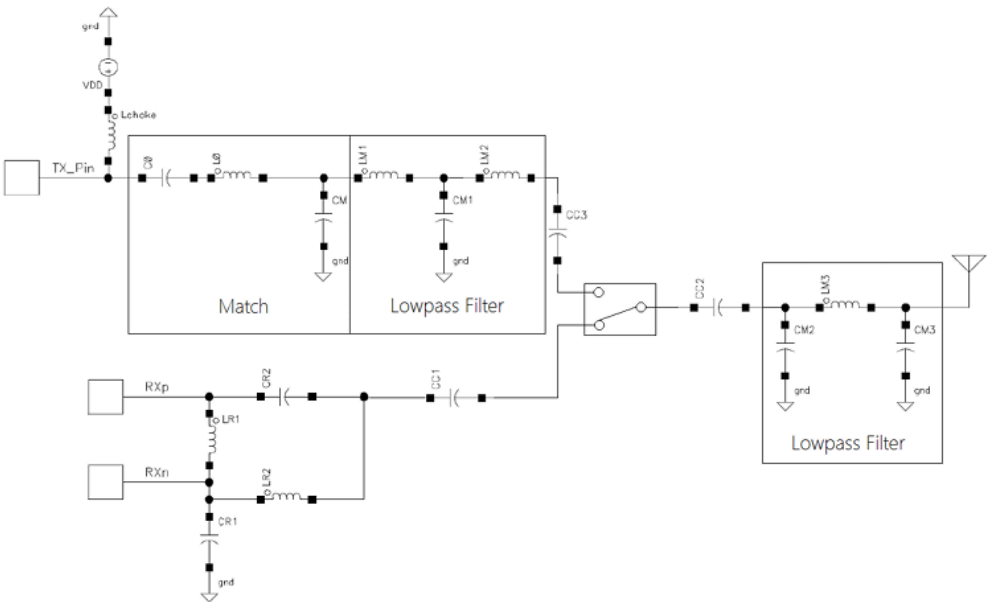


Figure 3. Class-E Matching Topology for Switched TX/RX Board Configuration

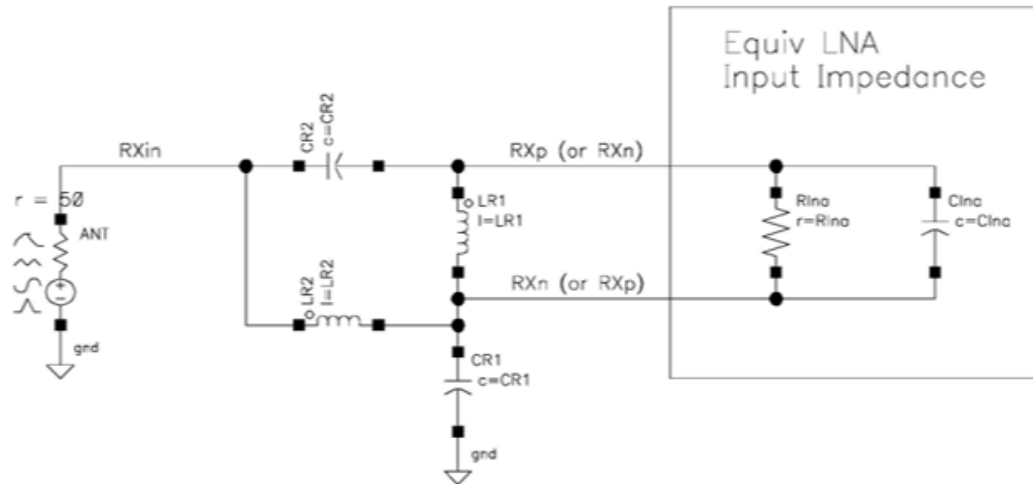
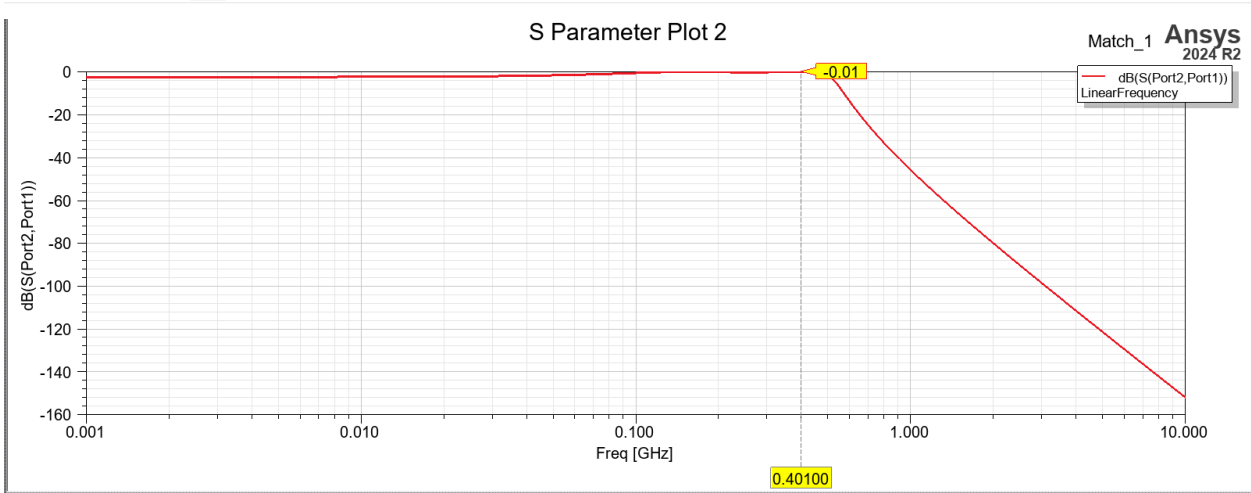
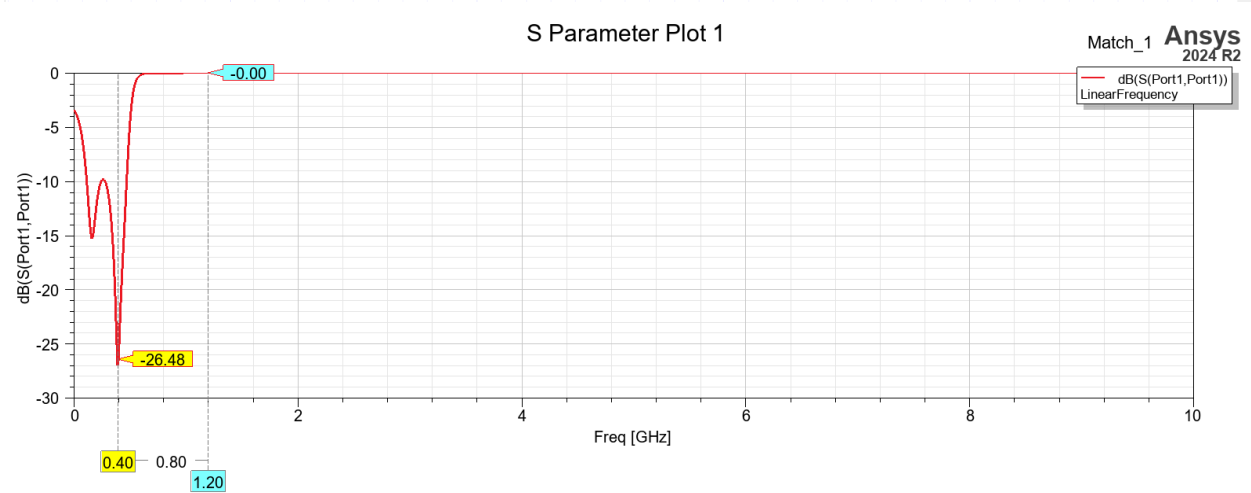
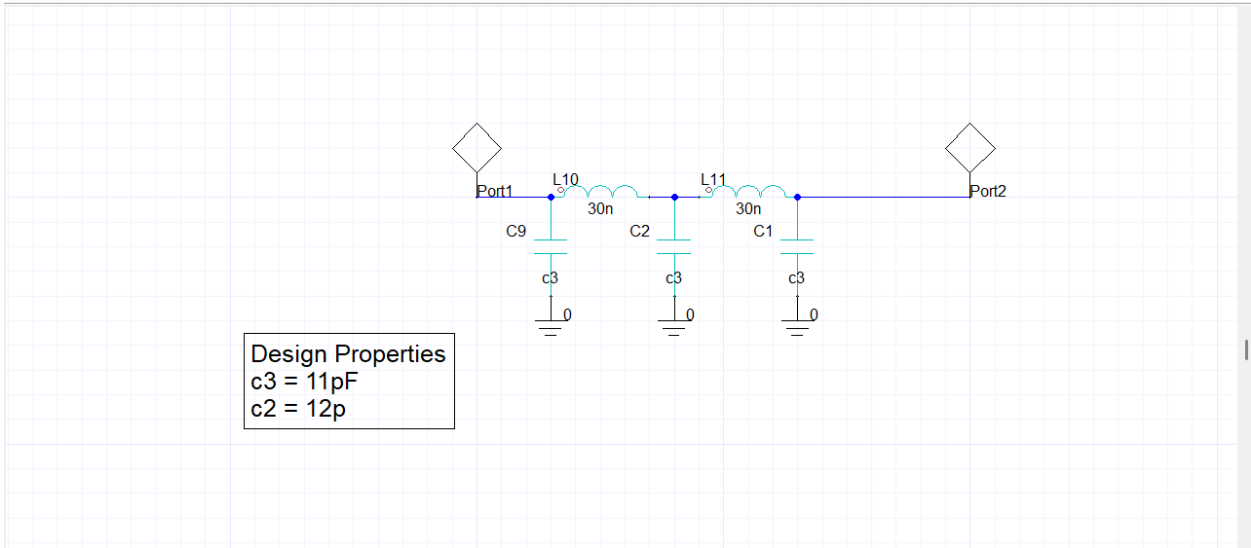


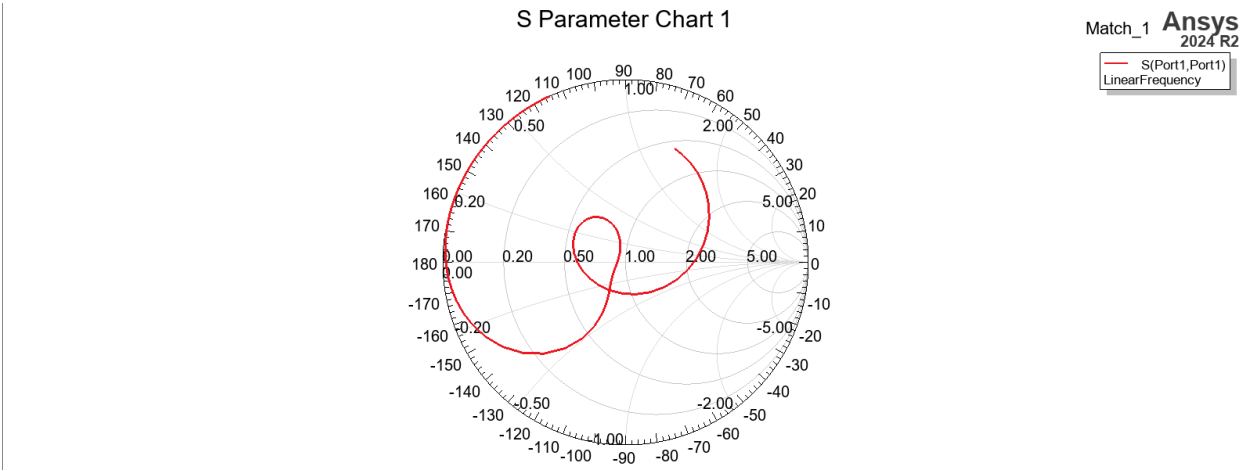
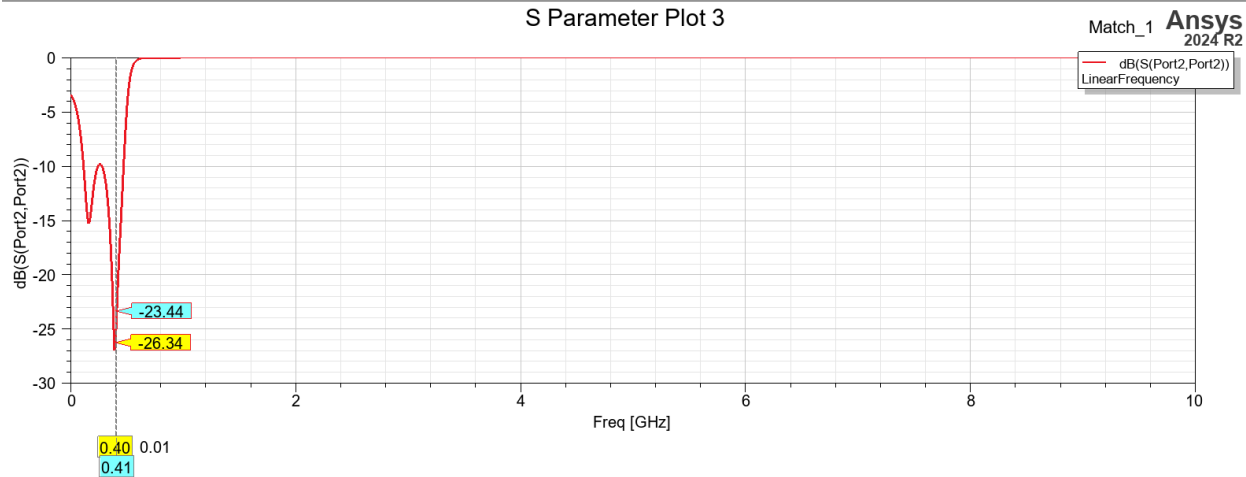
Figure 2. Four-Element Match Network

Table 4. 4-Element Match Network Component Values (Calculated)

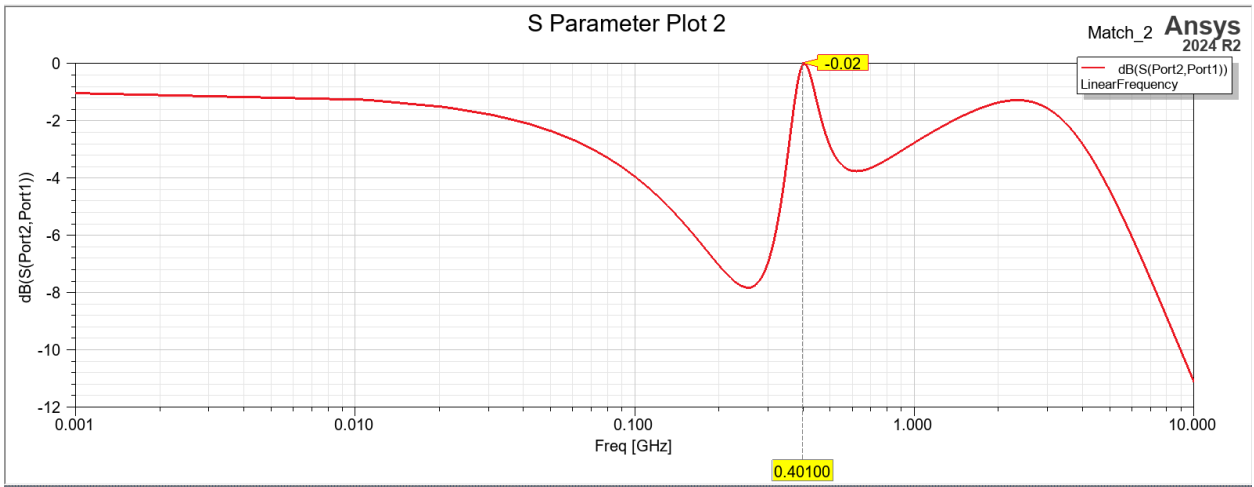
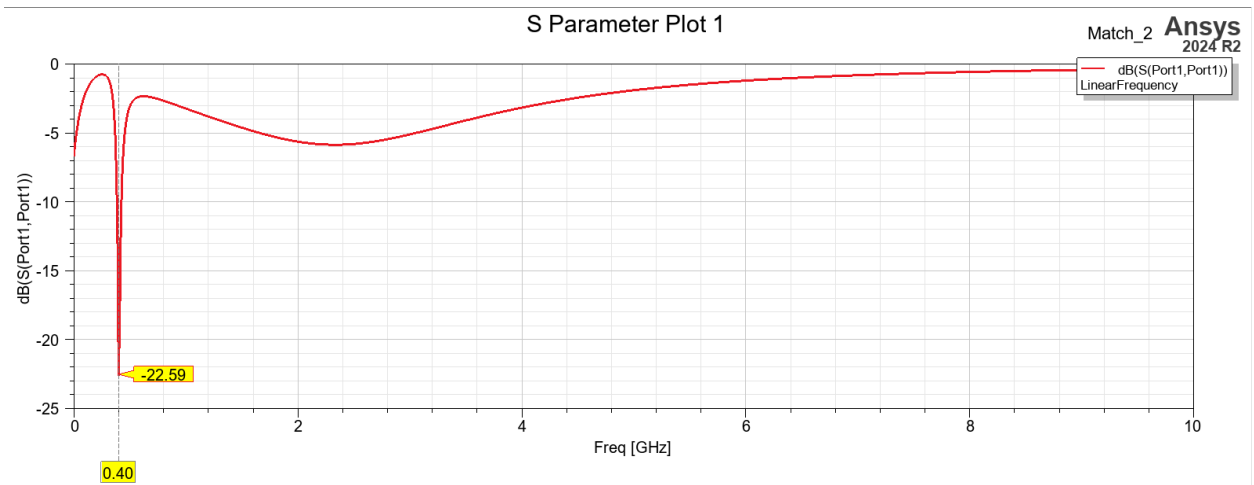
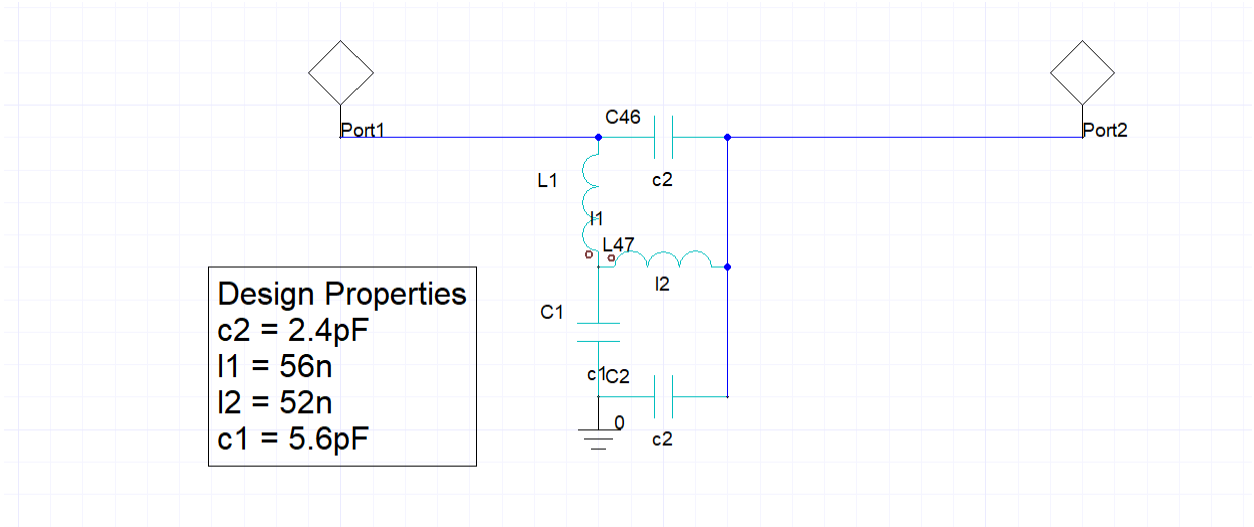
Freq	RLNA	CLNA	CR1	LR1	CR2	LR2
140 MHz	545 Ω	1.02 pF	13.77 pF	289.69 nH	6.89 pF	187.69 nH
169 MHz	536 Ω	0.98 pF	11.51 pF	230.20 nH	5.75 pF	154.15 nH
200 MHz	530 Ω	0.96 pF	9.78 pF	186.19 nH	4.89 pF	129.55 nH
250 MHz	520 Ω	0.94 pF	7.90 pF	138.84 nH	3.95 pF	102.63 nH
300 MHz	512 Ω	0.95 pF	6.63 pF	108.07 nH	3.32 pF	84.88 nH
315 MHz	509 Ω	0.95 pF	6.34 pF	100.91 nH	3.17 pF	80.58 nH
350 MHz	499 Ω	0.95 pF	5.76 pF	86.58 nH	2.88 pF	71.85 nH
390 MHz	491 Ω	0.96 pF	5.21 pF	73.63 nH	2.60 pF	63.96 nH
400 MHz	488 Ω	0.96 pF	5.09 pF	70.77 nH	2.55 pF	62.15 nH

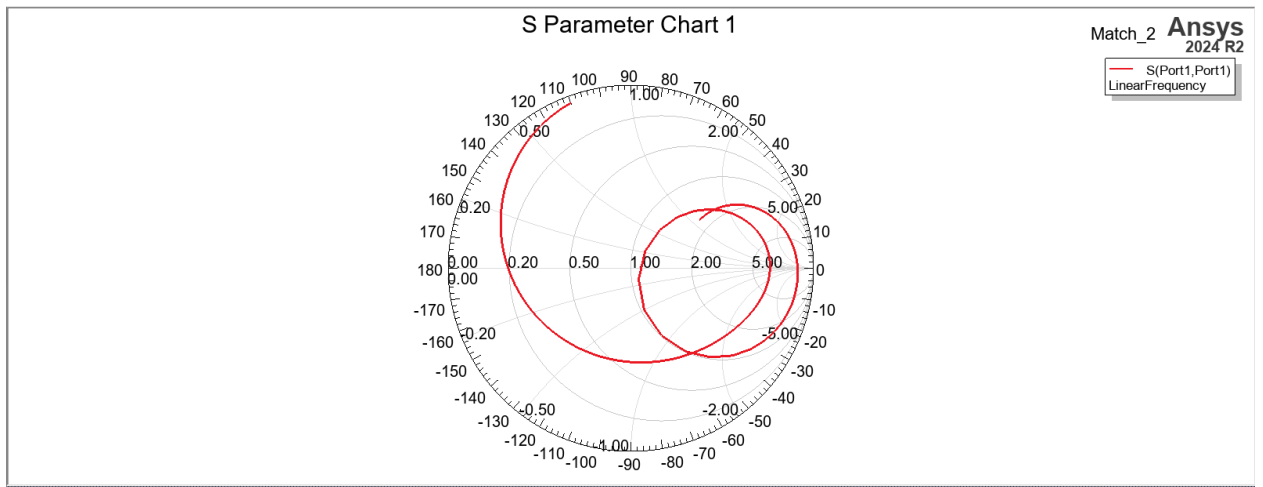
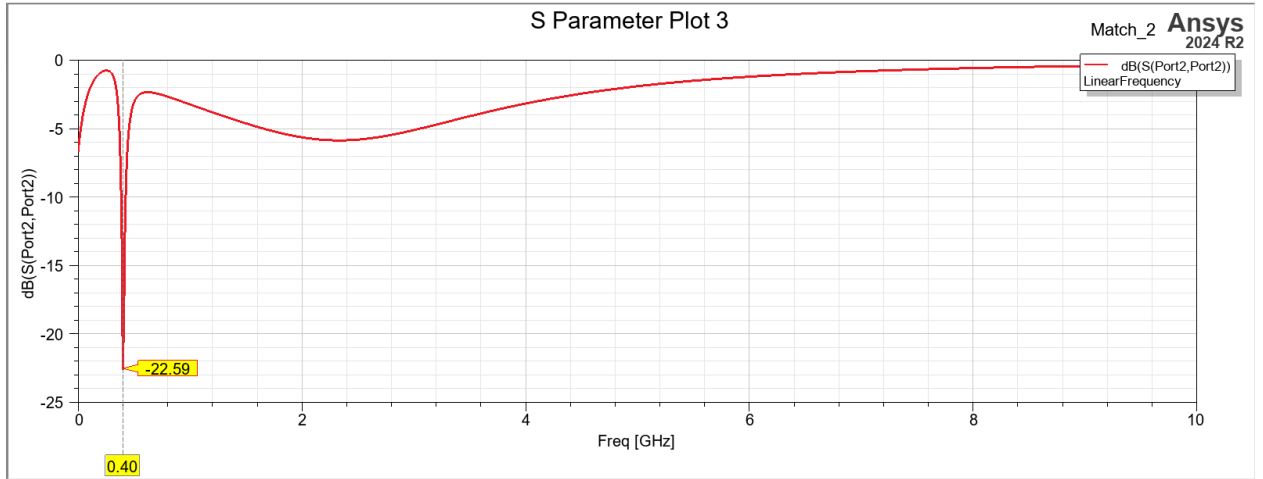
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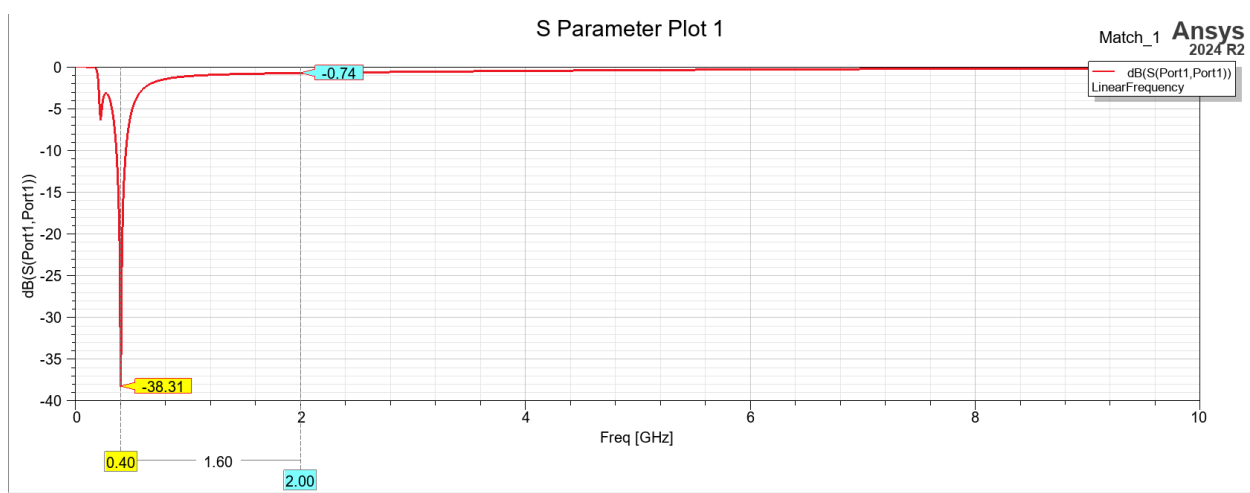
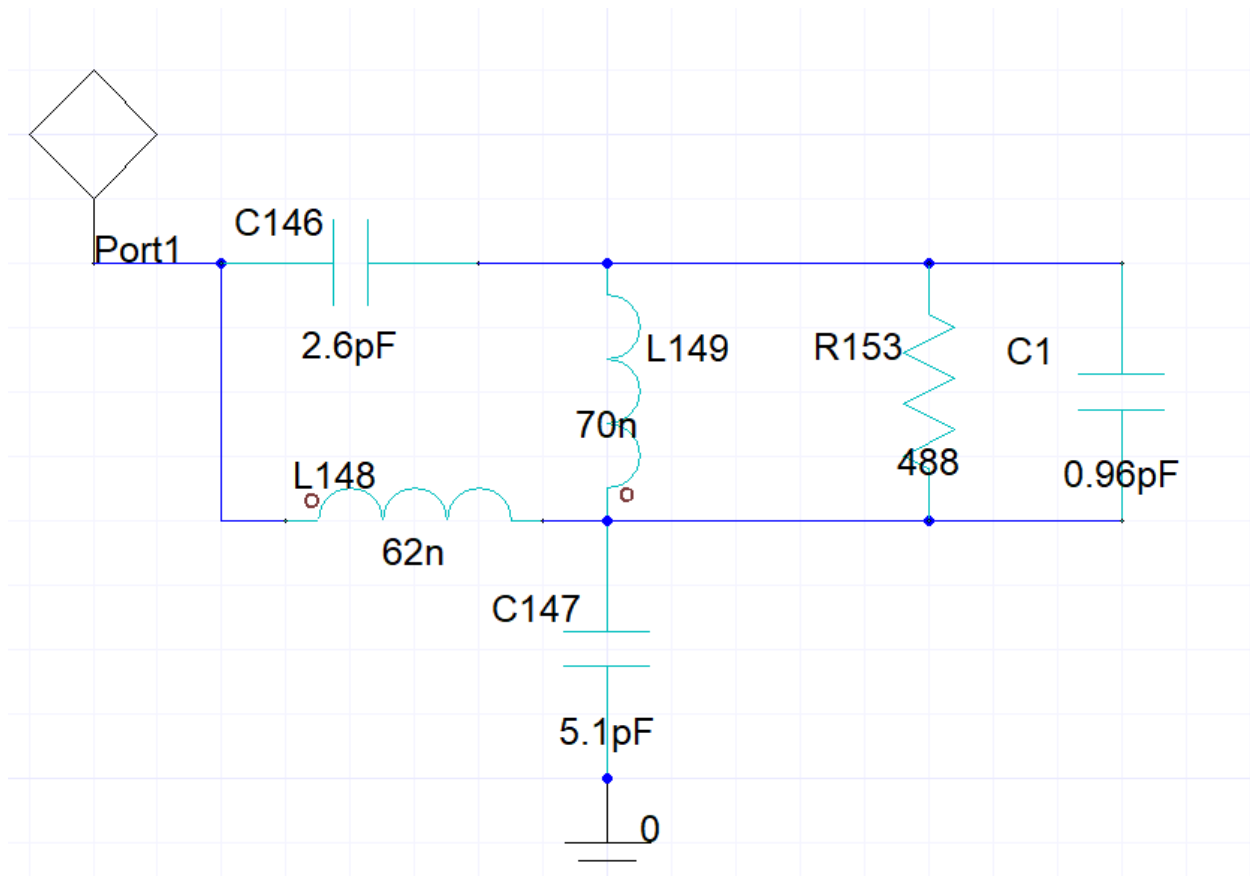


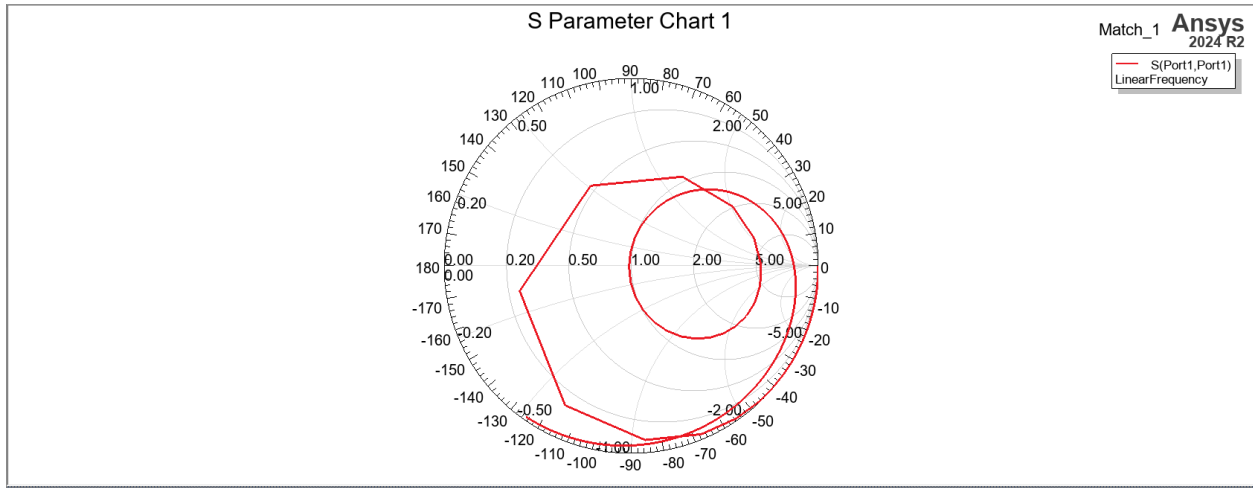
RX Match 2:





RX Match 3:





RX_PATH(simulated with simsurf components)